

SUPERVISED MULTI-OMICS ML FOR TARGET DISCOVERY IN FECD

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Base Paper & Motivation

- **Base Paper Reference:** *Zhang et al. (2023, PLOS ONE)* - Pooled 3 FECD RNA-seq datasets (GSE142538, PRJNA524323, GSE112201) and found RE+/RE- transcriptomes highly similar via DE & meta-analysis (edgeR/AWFisher). *No predictive model was built.*
- **Motivation:** FECD is a progressive, incurable corneal disease. Transcriptomic, epigenomic, and genomic layers have been studied individually, but never integrated through a predictive ML lens.

What is Already Solved & The Gap

- **Transcriptomics:** DE genes, splicing, and network analysis (association only).
- **Epigenomics:** CpG hyper/hypo-methylation identified in isolation.
- **Genomics:** GWAS loci + Polygenic Risk Scores (AUC 0.83) for genetic risk, not tissue prediction.
- **The Gap:** No supervised ML classifier on FECD tissue omics; no cross-layer integration framework.

Input and Output Definition

INPUTS

- RNA-seq ($34 \times \sim 20k$ genes)
- Methylation ($13-25 \times \sim 450k$ CpGs)
- GWAS/PRS (fixed prior list)



OUTPUTS

- Per-layer classifier (CV metrics)
- FECD vs Control (RE+/RE- exploratory)
- Ranked 3-6 gene shortlist (≥ 2 layers)

Dataset & Preprocessing Plan

Layer	Source Accessions	N Samples
Transcriptomics	GSE142538 / GSE112201	N = 34
Epigenomics	GSE94462 / GSE198917	N = 13-25
Genomics	FECD GWAS / PRS	Summary-level

PREPROCESSING PIPELINE:

QC → Normalization → Batch Correction (ComBat-seq) → In-fold Feature Scaling & Selection

Working RQ & Execution Timeline

"Can ML models trained independently on FECD transcriptomic and epigenomic profiles achieve cross-validated classification beyond chance, and do predictive features converge with GWAS evidence?"

PRES 1

Problem Setup, Base Paper & RQ

PHASE 2

Pipeline dummy run with small targeted genes.

PHASE 3

Data QC, Norm & ComBat-seq, Nested CV, Elastic Net + RF

PHASE 4

Convergence & Shortlist Report

Research Question, Scope & Deliverables

What Problem Are You Asking?

Can supervised ML models, trained independently on FECD transcriptomic and epigenomic profiles, achieve statistically meaningful cross-validated classification of disease/subtype status and do the resulting predictive features converge with independent genomic (GWAS) evidence to prioritize disease-relevant candidate genes for experimental validation?

Scope & Key Assumptions

- **Scope:** Primary task is binary FECD vs. Control classification. RE+/RE− subtype analysis is exploratory (due to N=4 constraint). Transcriptomics & epigenomics modeled independently; genomics enters as a fixed prior list. Integration occurs at the gene level.
- **Assumptions:**
 - Public data quality & sample annotations are sufficient.
 - ComBat-seq batch correction preserves biological signal.
 - Regularization (Elastic Net/RF) controls small-sample overfitting ($p \gg n$).
 - Cross-layer overlap is evaluated against a random-gene baseline.

Expected Contributions

- **Methodological Contribution:** First supervised machine learning classification framework applied to FECD tissue omics data.
- **Biological Discovery:** A multi-layer-supported candidate gene shortlist (3-6 genes) prioritized for wet-lab validation.
- **Methodological Benchmark:** A worked example of leakage-free, small-N genomics machine learning workflow.

Tentative Deliverables & Milestones

Milestone	Deliverable Summary
Presentation 1	Problem setup, base paper benchmarking, dataset plan, working RQ (this deck).
Midterm	Pipeline dummy run with small targeted genes.
Pre-Endterm	Data acquired & preprocessed; baseline per-layer models trained with CV results. Feature importance stability check; cross-omics convergence + pathway enrichment.
Endterm / Final	Final report, ranked candidate shortlist, and proposed wet-lab validation plan.